

**AI-BASED SMART EXPENSE TRACKING AND FINANCIAL
MANAGEMENT SYSTEM**

PHASE I PROJECT REPORT

Submitted by

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in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING



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BONAFIDE CERTIFICATE

Certified that this Report titled “**AI-BASED SMART EXPENSE TRACKING AND FINANCIAL MANAGEMENT SYSTEM**” is the bonafide work of **ABILASH G R-230701003, ABHISHEK ROBIN S A-230701005, ABISHEK I-230701009, SARVESHWAR P-2307010296** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

**“SDG 1 – No Poverty: Promotes financial planning and savings among families.
SDG 8 – Decent Work and Economic Growth: Encourages responsible financial management and economic well-being.
SDG 12 – Responsible Consumption and Production: Helps users identify unnecessary spending and develop responsible consumption habits.”**

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ABSTRACT

The proposed AI-Based Smart Expense Tracking and Financial Management System is a software-based platform developed to manage and monitor financial activities for families and employees. The system provides a parent-child money management feature, where parents can allocate a specific amount of money to children and monitor their expenses. Children can record their spending based on different categories, and the system automatically generates graphical reports such as pie charts to visualize expense patterns. The system also maintains employee salary information and monthly financial records for better income and expense management. The proposed system includes an intelligent high-expense detection feature that analyzes user spending patterns and identifies excessive expenses. When unusual or high spending is detected, the system can generate an automated email alert to notify the parent or authorized user. An OCR-based bill processing module is also incorporated, allowing users to upload bill or receipt images. The OCR technology extracts important information from the bill and helps automatically add the expense under the appropriate category, reducing manual data entry. In addition, an AI-based financial chatbot is developed using Natural Language Processing (NLP) to answer user queries related to expenses, budgeting, salary, and financial activities. A CNN-based model is used to support image-based bill classification and expense categorization. By integrating financial tracking, graphical analytics, OCR, AI-based expense detection, email alerts, NLP, and deep learning, the system provides an intelligent and user-friendly solution for effective financial monitoring and expense management.

ACKNOWLEDGMENT

Initially we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavor to put forth this report. Our sincere thanks to our Chairman **Mr. S. MEGANATHAN, B.E, F.I.E.**, our Vice Chairman **Mr. ABHAY SHANKAR MEGANATHAN, B.E., M.S.**, and our respected Chairperson **Dr. (Mrs.) THANGAM MEGANATHAN, Ph.D.**, for providing us with the requisite infrastructure and sincere endeavoring in educating us in their premier institution.

Our sincere thanks to **Dr. S.N. MURUGESAN, M.E., Ph.D.**, our beloved Principal for his kind support and facilities provided to complete our work in time. We express our sincere thanks to **Dr. J. Manoranjini,, Ph.D.**, Professor and Head of the Department of Computer Science and Engineering for his guidance and encouragement throughout the project work. We convey our sincere and deepest gratitude to our internal guide, **Dr. N. Duraimurugan**, Associate Professor, Department of Computer Science and Engineering for his valuable guidance throughout the course of the project. We thank our Project Coordinator, Mr. Karthik , B.E., M.E., Assistant Professor, Department of Computer Science and Engineering for his useful tips during our review to build our project.

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LIST OF ABBREVIATIONS

S. No.	Abbreviation	Expansion
1	AI	Artificial Intelligence
2	ML	Machine Learning
3	OCR	Optical Character Recognition
4	NLP	Natural Language Processing
5	CNN	Convolutional Neural Network
6	NER	Named Entity Recognition
7	NLTK	Natural Language Toolkit
8	BERT	Bidirectional Encoder Representations from Transformers
9	GPT	Generative Pre-trained Transformer
10	LSTM	Long Short-Term Memory
11	GUI	Graphical User Interface
12	DFD	Data Flow Diagram
13	UML	Unified Modeling Language
14	H/W	Hardware
15	S/W	Software
16	RAM	Random Access Memory
17	IDLE	Integrated Development and Learning Environment
18	PDF	Portable Document Format
19	OpenCV	Open Source Computer Vision Library
20	CNNs	Convolutional Neural Networks

CHAPTER 1

INTRODUCTION

1.1 GENERAL

The AI-Based Smart Expense Tracking and Financial Management System is an intelligent software solution designed to simplify and improve the management of personal, family, and employee financial activities. Traditional expense management methods often require manual recording and provide limited insights into spending behavior, making it difficult for users to monitor their financial activities effectively. The proposed system addresses these challenges by providing a centralized platform for recording, tracking, and analyzing income and expenses. It helps users understand their spending patterns through categorized expenses and graphical representations, while also supporting parent-child financial management and employee salary tracking. By incorporating artificial intelligence, OCR, NLP, and deep learning techniques, the system aims to provide a smarter, more efficient, and user-friendly approach to financial monitoring and management.

1.2 OBJECTIVE

The main objective of the proposed system is to develop an intelligent and user-friendly platform that enables users to efficiently track, manage, and analyze their financial activities. The system aims to automate expense recording through OCR-based bill processing and classify expenses into appropriate categories using AI techniques. It is designed to identify unusually high or excessive spending by analyzing expense patterns and provide timely email alerts to parents or authorized users. Another objective is to provide graphical reports that help users understand their spending behavior and maintain better financial control. The system also aims to provide an NLP-based financial chatbot for answering user queries related to

expenses, salary, budgeting, and financial records. Overall, the objective is to reduce manual financial management and support better financial decision-making through AI, OCR, NLP, and deep learning technologies.

1.3 EXISTING SYSTEM

The existing system relies mainly on notebooks, spreadsheets, calculators or basic expense-tracking applications. Users manually record expenses, categorize them and enter bill details. Such systems generally lack intelligent spending analysis, integrated parent-child monitoring, automated bill extraction, high-expense detection, alerts and conversational financial assistance.

The disadvantages specifically identified include time-consuming manual entry, data-entry errors, difficulty in identifying unusual expenditure, manual bill processing, limited financial analysis, lack of automatic notifications and absence of an AI financial chatbot.

1.4 PROPOSED SYSTEM

The existing system relies mainly on notebooks, spreadsheets, calculators or basic expense-tracking applications. Users manually record expenses, categorize them and enter bill details. Such systems generally lack intelligent spending analysis, integrated parent-child monitoring, automated bill extraction, high-expense detection, alerts and conversational financial assistance.

The disadvantages specifically identified include time-consuming manual entry, data-entry errors, difficulty in identifying unusual expenditure, manual bill processing, limited financial analysis, lack of automatic notifications and absence of an AI financial chatbot.

CHAPTER 2

LITERATURE SURVEY

This section of the paper, provides an all-round review about multiple existing systems that holds similar objective and problem statements as that of the proposed project. The study of these existing papers will produce valuable insights for the construction of the proposed system.

2.1 LITERATURE REVIEW

Paper 1 – A Conversation-Driven Approach for Chatbot Management

Santos et al. proposed a **Chatbot Management Process (CMP)** for improving a chatbot continuously after deployment. The process is based on analysing real user conversations and updating the chatbot knowledge base through a human-supervised cycle.

The method was evaluated using the EvaTalk chatbot. The study reported a reduction of human hand-off from 44.43% to 30.16% and an increase of about 160% in knowledge-base examples.

This paper supports the chatbot component of the Expense Tracking System because financial chatbot responses can similarly be improved by analysing user interactions.

Paper 2 – Assessing the Impact of Chatbot-Human Personality Congruence

Kuhail et al. studied whether the personality of a chatbot affects user trust, engagement and willingness to use the chatbot.

The study involved 54 college students interacting with three types of chatbots:

- extroverted;
- agreeable;

- conscientious.

The findings showed stronger personality-congruence effects among extroverted participants, while the results for agreeable and conscientious participants were not conclusive.

This paper is relevant to the design of a more user-friendly financial chatbot.

Paper 3 – Design of a Personal Credit Risk Prediction Model and Legal Prevention of Financial Risks

Zhuang and Wei developed a financial-risk prediction system using **GAN/WGAN-GP and LightGBM**.

GAN was used to generate additional minority-class samples and solve the problem of imbalanced financial datasets. LightGBM was then applied to classification.

The study reported approximately 86.7% average classification accuracy in one comparison and strong performance using AUC, F1-score, Recall and KS metrics.

The paper is useful for understanding financial-pattern analysis and abnormal-behaviour detection.

Paper 4 – Systemic Financial Risk Measurement Based on HMM and Text Mining

Li, Cai and Hu proposed a financial-risk measurement method combining traditional financial information with internet-search information.

The researchers used:

- Hidden Markov Model;
- text mining;
- financial indicators;

- Baidu search information.

The study concluded that combining structured financial information with online information improved systemic-risk measurement.

This supports the general concept of analysing multiple financial signals for intelligent financial monitoring.

Paper 5 – Financial Statement Fraud Detection Using Deep Learning

Xiuguo and Shengyong developed a fraud-detection framework using both numerical and textual financial information.

The study used:

- Word2Vec;
- CNN;
- LSTM;
- GRU;
- financial indicators;
- non-financial indicators;
- MD&A textual information.

The reported classification accuracy was approximately 94.98% for LSTM and 94.62% for GRU.

This paper demonstrates how financial information can be preprocessed, converted into features and classified using deep-learning methods.

2.2 COMPARISON AND DISCUSSION

The literature reviewed demonstrates the application of different Artificial Intelligence and Machine Learning techniques in chatbot development, financial risk analysis, fraud detection, and intelligent financial management.

The first paper, **“A Conversation-Driven Approach for Chatbot Management,”** focuses on improving chatbot performance after deployment. It introduces a Chatbot Management Process in which user conversations are continuously analysed and used to update the knowledge base. This approach is relevant to the proposed Expense Tracking System because the financial chatbot can be improved based on the questions and interactions of users.

The second paper, **“Assessing the Impact of Chatbot-Human Personality Congruence on User Behavior,”** examines how chatbot personality influences user trust, engagement, and usage intention. The study shows that the communication style of a chatbot can affect user experience. This concept is useful for designing a financial chatbot that is simple, interactive, and comfortable for users.

The third paper, **“Design of a Personal Credit Risk Prediction Model and Legal Prevention of Financial Risks,”** uses GAN and LightGBM techniques for financial risk prediction. GAN is used to overcome the problem of imbalanced financial datasets, while LightGBM is used for classification. Although the paper mainly deals with credit risk, its methods are useful for understanding abnormal financial behaviour and intelligent financial analysis.

The fourth paper, **“Research on Systemic Financial Risk Measurement Based on HMM and Text Mining,”** combines traditional financial information with internet-based information to identify changes in financial risk. The study demonstrates that intelligent analysis of different types of financial data can improve

risk monitoring. This concept supports the proposed system's objective of analysing expense patterns and identifying unusual or excessive expenditure.

The fifth paper, “**An Analysis on Financial Statement Fraud Detection for Chinese Listed Companies Using Deep Learning,**” combines numerical financial data and textual information using deep learning methods such as CNN, LSTM, GRU, and Word2Vec. The study demonstrates the importance of preprocessing, feature extraction, and classification in financial-data analysis. These techniques are relevant to the proposed Expense Tracking System for expense categorization and intelligent financial analysis.

A comparison of the reviewed papers is given below:

Paper	Technique Used	Main Purpose	Relevance to Proposed System
Paper 1	Chatbot Management, NLP	Improve chatbot performance	Financial chatbot improvement
Paper 2	Personality-Based Chatbot	Improve trust and engagement	Better user interaction
Paper 3	GAN, WGAN-GP, LightGBM	Credit-risk prediction	Financial pattern and anomaly analysis

Paper 4	HMM, Text Mining	Financial-risk measurement	Intelligent financial monitoring
Paper 5	CNN, LSTM, GRU, Word2Vec	Financial-fraud detection	Financial-data classification

From the above comparison, it can be observed that the selected research works concentrate on different aspects of intelligent financial systems. Chatbot-related studies support the development of interactive financial assistance, while financial-risk and fraud-detection studies provide useful concepts for preprocessing, classification, and abnormal-pattern detection.

The proposed **AI-Based Smart Expense Tracking and Financial Management System** combines these concepts in a single application by integrating expense tracking, OCR-based bill processing, intelligent expense categorization, high-expense detection, graphical analysis, automated alerts, and an NLP-based financial chatbot.

2.3 CONCLUSION

The literature survey shows that Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing, and automated data analysis have become important techniques in the development of intelligent financial and chatbot-based systems. The reviewed papers demonstrate that these technologies can improve the accuracy of financial analysis, automate repetitive tasks, support user interaction, and provide better decision-making assistance.

The chatbot-related studies show that the performance of a chatbot can be improved by analysing user conversations, updating the knowledge base, and providing a more

suitable communication style. These concepts are useful for the proposed financial chatbot, which is intended to answer user queries related to expenses, salary, budgeting, and savings.

The financial risk and fraud detection studies demonstrate the importance of data preprocessing, feature extraction, classification, and abnormal-pattern detection. Techniques such as GAN, LightGBM, HMM, CNN, LSTM, GRU, Word2Vec, and text mining provide useful approaches for analysing financial information and identifying unusual behaviour.

The reviewed studies also indicate that combining different types of data can improve the performance of intelligent systems. Numerical financial information, textual information, transaction records, and user interaction data can all contribute to more effective financial monitoring and analysis.

Another important observation from the literature is that intelligent systems should not depend only on prediction accuracy. User trust, system usability, continuous model improvement, and proper handling of data are also important for the successful operation of an AI-based financial system.

Based on these studies, the proposed AI-Based Smart Expense Tracking and Financial Management System combines several useful concepts into a single platform. The system uses OCR for bill processing, AI techniques for expense categorization and high-expense detection, graphical analysis for financial monitoring, automated alerts for unusual spending, and an NLP-based chatbot for user assistance.

Thus, the literature survey provides a strong technical foundation for the proposed system and supports the selection of suitable methods for financial-data processing, intelligent classification, user interaction, and automated expense management.

CHAPTER 3

SYSTEM DESIGN

This chapter of the paper, briefly discusses about the design of the proposed system with the help of Model diagrams. The requirements of the system and deployment are also extensively studied. This chapter also covers the overall architecture of the proposed system along with the overview of modules involved.

3.1 SYSTEM ARCHITECTURE

The system architecture of the AI-Based Smart Expense Tracking and Financial Management System is designed as an integrated structure in which different users and functional modules communicate through a common financial-management platform. The architecture mainly connects the Parent Module, Child Module, Employee Module, Expense Management Module, OCR and CNN-Based Bill Processing Module, AI Expense Detection Module, Financial Chatbot, and the database. The purpose of this architecture is to provide a centralized system for recording financial transactions, processing bills, analysing expenses, monitoring spending behaviour, and assisting users through an intelligent chatbot. The proposed system therefore combines normal financial-management functions with Artificial Intelligence, Optical Character Recognition, Natural Language Processing, and deep-learning techniques.

The first stage of the architecture is the user access and dashboard section. A user enters the application through the registration or login process. After successful authentication, the user is directed to the dashboard. The dashboard acts as the main interface through which the available financial-management functions can be accessed. Depending on the type of user, the system provides functions related to parent-child money management, employee financial records, expense entry, bill

processing, financial analysis, and chatbot interaction. This centralized design allows financial information to be maintained within a single application instead of using separate systems for different activities. The proposed system is specifically intended to provide centralized financial monitoring for parents, children, and employees.

The Parent Module forms an important part of the architecture. A parent can create and manage child accounts and allocate a specific amount of money to a child. After the money is allocated, the parent can monitor how the child spends the amount. The expense details recorded by the child are stored by the system and can later be viewed by the parent. This arrangement provides better control over parent-child financial activities and helps parents understand the spending behaviour of their children. The architecture therefore connects the parent interface with money allocation, child expense records, graphical analysis, and high-expense monitoring functions. The project source identifies the Parent Module and Child Module as separate modules within the overall system.

The Child Module allows the child user to record day-to-day expenses under different categories. After receiving the allocated amount, the child can enter spending details whenever an expense occurs. These expense records are stored as financial transactions and are made available for further analysis. The system can present the recorded information graphically, including through pie charts, so that the distribution of expenditure can be easily understood. This also enables the parent to identify how the allocated amount has been spent across different categories. The proposed system specifically provides categorized expense recording and graphical representation of spending patterns.

The Employee Module is included to maintain salary and monthly financial information. Employee users can maintain income and expenditure records in the

same system. Monthly salary details can be stored and compared with expenses to help users understand their financial position. By integrating employee salary management with the expense-management functions, the system avoids the need to maintain salary information and daily expense information separately. The source states that employee salary details are maintained on a monthly basis so that income and expenses can be monitored efficiently.

The Expense Management Module acts as the central financial-processing component of the architecture. Whenever a user adds an expense, the transaction information is recorded and stored. The system organizes expenses according to different categories and maintains the financial transaction history. The stored information can subsequently be analysed to identify spending patterns. Graphical representations are also generated to provide users with a clearer understanding of their expenditure. Therefore, this module connects user-entered financial information with the database, expense analysis, graphical reporting, and the high-expense detection mechanism. The overall project is intended to provide a single platform for managing expenses, income, salary details, and financial transactions.

Another important component is the OCR-Based Bill Processing Module. Instead of manually entering all the information available on a bill or receipt, the user can upload an image of the bill. Optical Character Recognition is then used to extract important information from the uploaded image. The source specifically mentions extraction of details such as the bill amount and other bill information. The extracted information can then be used for expense recording, thereby reducing the amount of manual data entry required from the user. This is particularly useful when users have several receipts or bills that must be entered into the system.

The architecture also includes a CNN-Based Classification component. After the bill information has been extracted through OCR, a CNN-based image classification

model assists in identifying the appropriate expense category. Thus, OCR performs the information extraction function, while the classification component helps determine the type of expense. The combination of these two techniques supports automated bill processing and improves the process of categorizing expenditure. The project source specifically states that OCR extracts important information from bill or receipt images while CNN-based image classification assists in identifying the appropriate expense category.

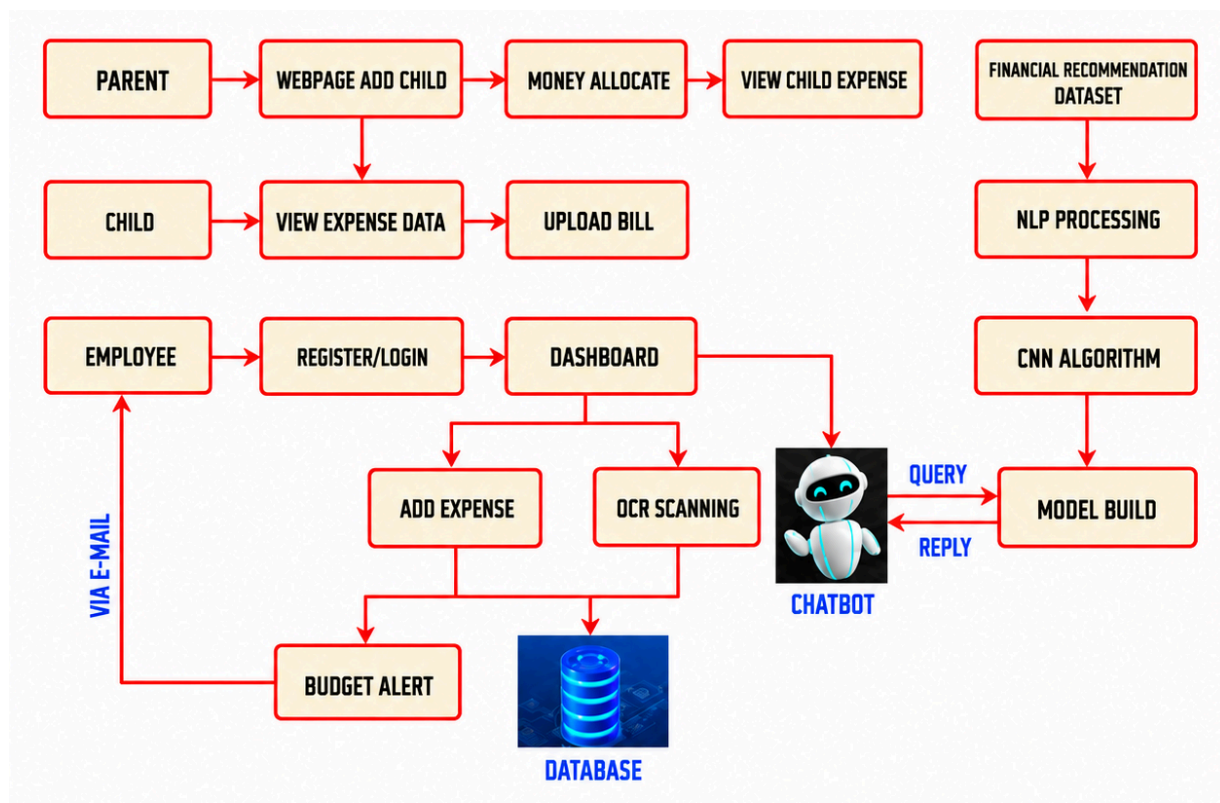
The database functions as the central storage component of the system. Financial transactions created through manual expense entry or bill processing are stored so that they can be retrieved for subsequent operations. The database supports functions such as viewing child expenses, maintaining salary information, analysing financial transactions, generating graphical representations, identifying high expenditure, and responding to user queries through the chatbot. In this way, the different modules of the system are interconnected through common financial information rather than functioning independently.

The AI-Based High-Expense Detection Module analyses user expenditure to identify excessive or unusual spending. The proposed system allows a financial limit to be defined. When the recorded expenses exceed this defined limit, the system identifies the condition as high expenditure. An automatic email notification can then be generated and sent to the parent or other authorized user. This provides users with timely information regarding unusual spending and helps them take corrective financial action. The project source specifically provides for AI-based expense monitoring and automatic email alerts when expenses exceed the defined financial limit.

The Financial Chatbot Module provides conversational assistance to users. The chatbot uses Natural Language Processing to understand queries relating to expenses,

salary, budgeting, savings, and other financial activities. A user can submit a financial question through the interface, and the system processes the query before generating a relevant response. This allows users to interact with their financial records through natural-language questions rather than relying only on menus and reports. The chatbot therefore provides an additional intelligent interface between the user and the financial-management system.

The overall architecture can be understood through three major flows. In the general expense flow, the user proceeds through **Register/Login** → **Dashboard** → **Add Expense** → **Bill Upload** → **OCR Scanning** → **Expense Categorization** → **Database** → **Expense Analysis** → **Budget Alert**. In the parent-child financial flow, the process is **Parent** → **Child Management** → **Money Allocation** → **Child Expense Recording** → **View Child Expense** → **Expense Analysis**. In the AI chatbot flow, financial information and user queries are processed through the NLP and AI components before the system produces an appropriate chatbot response.



Thus, the architecture integrates user management, financial-data entry, parent-child monitoring, employee salary management, automated bill processing, AI-based expense categorization, expense analysis, alert generation, and conversational financial assistance into a single system. The architecture is designed to reduce manual financial-management work, provide better visibility of spending patterns, automate routine operations, and make financial information easier for users to understand and manage. The source describes the proposed system as combining parent-child money allocation, employee salary management, expense analytics, high-expense alerts, OCR-based bill processing, CNN-based classification, and an NLP-based financial chatbot in one intelligent and user-friendly platform.

3.2 SYSTEM REQUIREMENTS

This section describes the requirements of the proposed system. Various aspects of the system like the hardware and software compatibility, the data requirements, dependencies and other ethical requirements are discussed in detail.

3.2.1 SOFTWARE REQUIREMENTS

The proposed system requires a suitable software environment to support expense tracking, bill processing, financial analysis, and chatbot functions. Python is used as the main programming language, along with OCR, image-processing, NLP, and deep-learning tools for implementing the intelligent features of the system. The required software components and their purposes are listed in the following table.

S. No.	Software / Technology	Specification / Purpose
1	Operating System	Windows 10 / Windows 11
2	Programming Language	Python
3	Development Framework	Python GUI Framework
4	Development Environment	IDLE
5	Python Version	Python 3.10.8
6	Tesseract OCR	Used for extracting text and financial information from uploaded bills and receipts
7	EasyOCR	Used as an OCR library for recognizing text from bill and receipt images
8	OpenCV	Used for image processing and preprocessing of uploaded bill images
9	NLTK	Used for Natural Language Processing operations in the financial chatbot
10	SpaCy	Used for processing and understanding user queries

11	Word2Vec / GloVe	Used for representing text in numerical vector form for NLP processing
12	CNN	Used to support classification of bills and expense categories
13	NLP	Used to understand financial queries related to expenses, salary, budgeting, and savings

3.2.2 HARDWARE REQUIREMENTS

The proposed system requires basic hardware resources to support smooth execution of the application, financial data processing, OCR-based bill recognition, and AI-related operations. A suitable processor, sufficient memory, and adequate storage space are necessary for reliable system performance. The minimum hardware requirements are given in the following table.

S. No.	Hardware Component	Requirement
1	Processor	Intel Core i3 / i5 / i7 or AMD Processor
2	RAM	Minimum 8 GB or above

3	Hard Disk	500 GB
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3.2.3 DATASET REQUIREMENTS

The project material does **not specify a fixed dataset size**.

It does state that the chatbot can use a **custom labelled sentence dataset**, with examples such as expense queries and budget-alert requests.

The system also processes:

- expense transaction information;
- salary information;
- bill and receipt images;
- expense categories;
- chatbot query text.

For example, the CNN material describes labelled sentences such as expense-query intents used to classify the user's request.

Therefore, the project dataset can be described as consisting of **financial transaction records, bill/receipt images and labelled chatbot-query data**, but the source does not provide an exact number of samples.

3.2.4 DEPLOYMENT AND SCALING

The proposed system requires suitable hardware resources to support the centralized management of income and expenses, parent-child financial monitoring, employee salary records, graphical expense analysis, and the processing of financial transactions.

The system also incorporates OCR-based bill and receipt processing, AI-based high-expense detection, CNN-based expense categorization, automated email alerts, and an NLP-based financial chatbot. To execute these functions efficiently, the project specifies a processor such as Intel i3, i5, i7 or AMD, a minimum of 8 GB RAM or above, and 500 GB of hard disk storage.

3.3 SYSTEM DESIGN

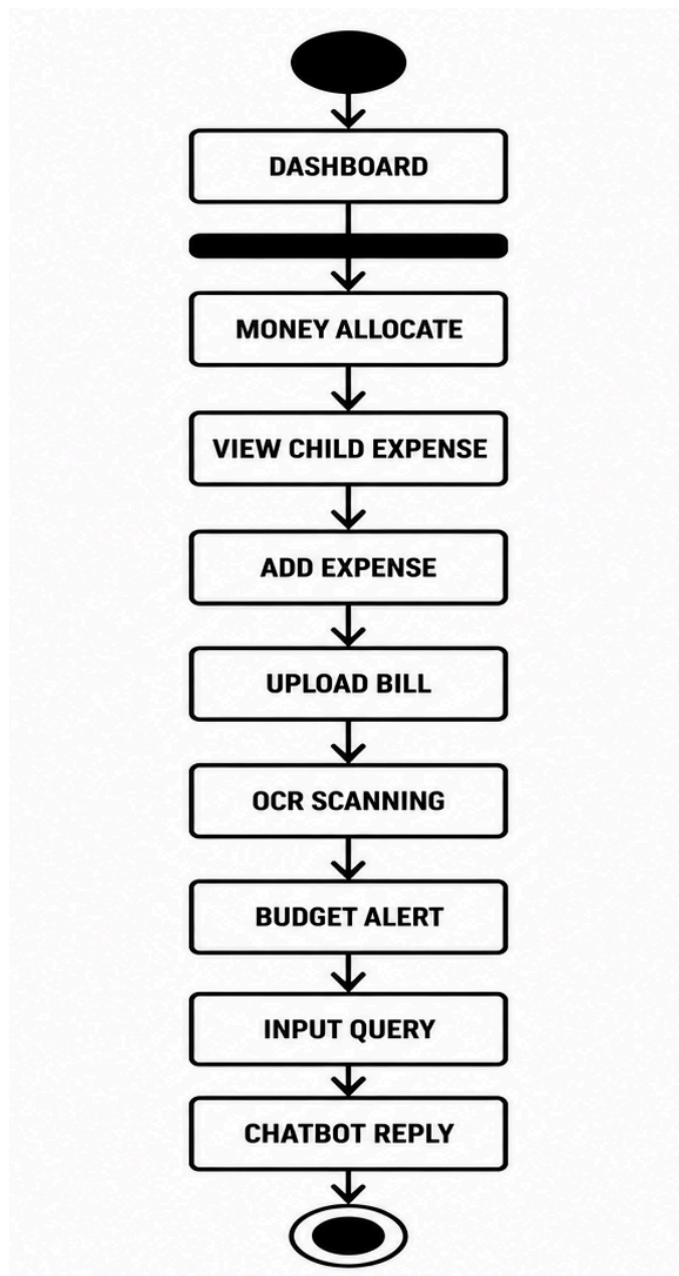
The **AI-Based Smart Expense Tracking and Financial Management System** is a centralized platform for recording, tracking, and analysing financial activities. It supports parent-child financial management, employee salary records, expense categorization, graphical reports, OCR-based bill processing, high-expense detection, email alerts, and an NLP-based financial chatbot.

The system includes six main modules: Parent, Child, Employee, Expense Management, OCR & CNN-Based Bill Processing, and AI Expense Detection & Financial Chatbot. The Parent Module manages child accounts, allocates money, monitors expenses, sets spending limits, and receives alerts. The Child Module records expenses and uploads bills or receipts. The Employee Module maintains salary and monthly expense details. The Expense Management Module stores, categorizes, and analyses transactions using graphical reports.

The OCR module extracts bill details, such as the amount, from uploaded receipts, while the CNN model helps classify the expense category. The AI module detects excessive or unusual spending and sends alerts to the parent or authorized user. The NLP chatbot answers questions related to expenses, salary, budgeting, and savings.

3.3.1 ACTIVITY DIAGRAM

The activity diagram represents the step-by-step workflow of the proposed system. It shows how the user moves through the main functions such as the dashboard, money allocation, viewing child expenses, adding expenses, uploading bills, OCR scanning, budget alerts, entering queries, and receiving chatbot responses. It provides a simple view of how different activities are performed in sequence within the system.



3.3.2 DATA FLOW DIAGRAM

The Data Flow Diagram represents how financial information moves through the proposed system from one process to another. It shows the flow of data between the dashboard, money allocation, child expense monitoring, expense entry, bill upload, OCR scanning, budget alert, user query, and chatbot response. The diagram helps to understand how different system functions are connected and how financial data is processed at each level.

LEVEL:0

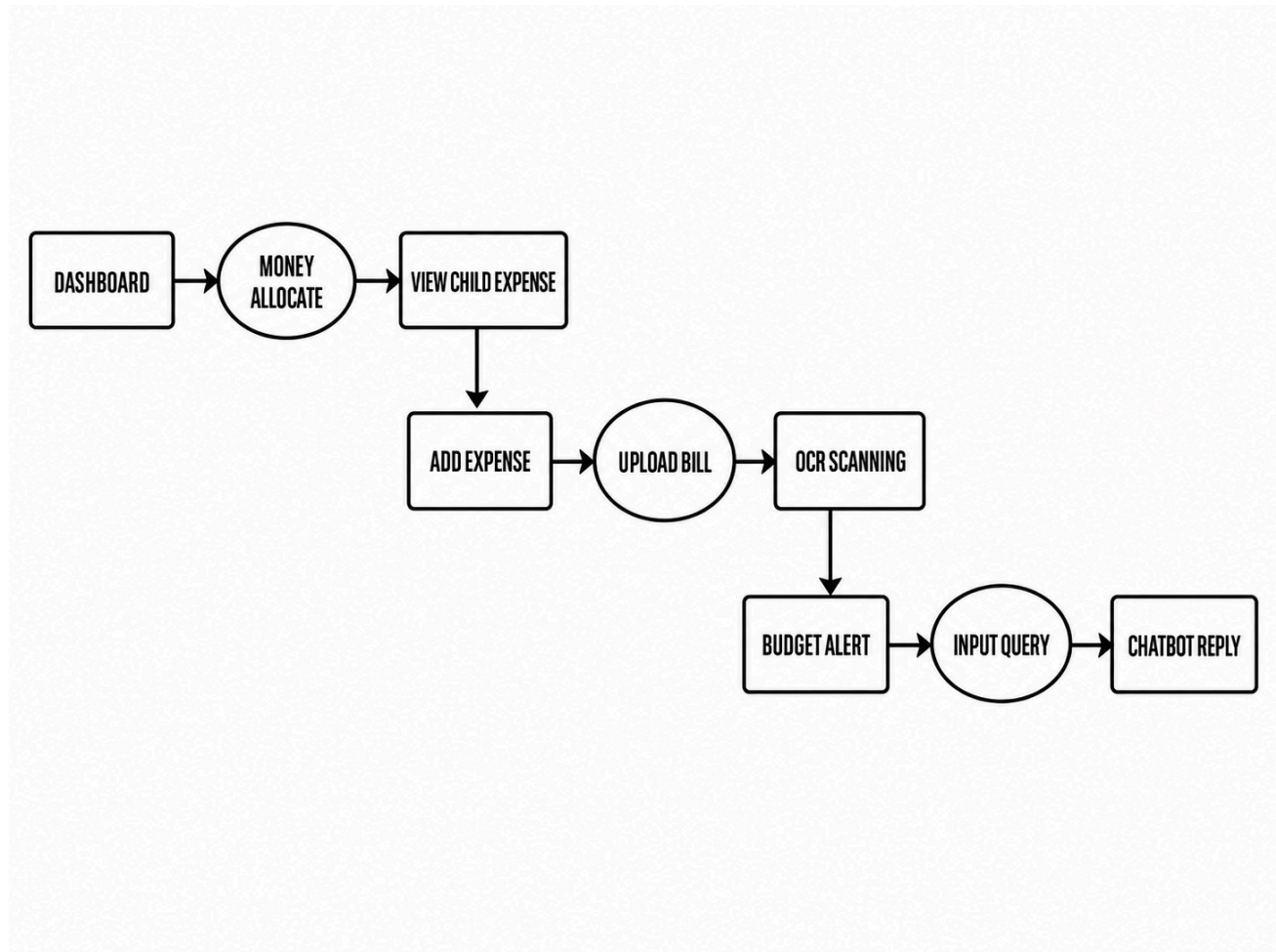


LEVEL:1



LEVEL:2





CHAPTER 4

PROJECT DESCRIPTION

4.1 METHODOLOGIES

The proposed system mainly uses three intelligent methodologies.

Optical Character Recognition

OCR extracts information from uploaded receipts and bills. It converts receipt images into machine-readable information such as:

- amount;
- date;

- merchant details;
- product information.

Tesseract OCR, EasyOCR and OpenCV are identified as suitable technologies.

Natural Language Processing

NLP enables the financial chatbot to understand user questions.

The source identifies:

- tokenization;
- Named Entity Recognition;
- intent recognition;
- optional sentiment analysis;
- stop-word removal;
- text normalization;
- Word2Vec/GloVe;
- NLTK/SpaCy.

Convolutional Neural Network

CNN is used for classification.

Its basic components include:

- embedding layer;
- convolutional layer;
- pooling layer;
- fully connected layer;
- Softmax output layer.

4.2 MODULES

The proposed AI-Based Smart Expense Tracking and Financial Management System is divided into different functional modules for managing user information, financial transactions, bill processing, expense analysis, alerts, and chatbot interaction. The modules work together to provide centralized financial management for parents, children, and employees. The original report includes separate modules for parent management, child expense recording, employee salary management, expense management, OCR and CNN-based bill processing, and AI-based expense detection with a financial chatbot.

4.2.1 DATA MANAGER PACKAGE

The Data Manager Package can contain the financial-data management functions described in the Expense Management Module of the original report. This component is responsible for recording, categorizing, storing, and analysing financial transactions. Users can manually enter expenses and assign them to appropriate categories. The system maintains transaction history and generates graphical reports such as pie charts to display spending patterns. It also supports comparison between income and expenditure and helps users monitor their daily and monthly financial activities.

The package can also include employee financial records because the Employee Module maintains salary information, monthly income, and expense details. These records can be categorized, stored, reviewed, and represented graphically to understand the relationship between income and expenditure.

4.2.2 CORE PACKAGE

The Core Package can represent the principal financial-management functions of the system, particularly the Parent, Child, and Employee Modules. The Parent Module

allows parents to create and manage child accounts, allocate money, monitor child expenses, review spending patterns through graphical reports, define spending limits, and receive automated email alerts when excessive spending is detected.

The Child Module enables children to record their daily expenses within the amount allocated by their parents. Expenses can be entered under categories such as food, travel, education, and shopping. The system stores these transactions and provides graphical representations to help children understand their spending habits. Children can also upload bills or receipts for automated processing.

The Employee Module maintains salary and monthly financial records and allows users to monitor their income and expenditure in an organized manner.

4.2.3 MODERATION PACKAGE

The uploaded report does not contain a separate module called Moderation Package. The closest source-supported function is the high-expense monitoring and alert mechanism.

The AI Expense Detection component analyses transaction data and identifies excessive or unusual spending based on predefined limits and spending patterns. When high expenditure is detected, the system automatically generates an email alert to the parent or authorized user. This function helps users monitor financial activities and take action when spending exceeds the specified limit.

Therefore, if the heading “Moderation Package” must remain in the prescribed format, it can be used for the system's expense-monitoring and alert-control functions, but this is a reorganization of the original report rather than an original module name.

4.2.4 REINFORCEMENT PACKAGE

The uploaded report also does not contain a separate Reinforcement Package. The closest relevant functions are the AI Expense Detection and Financial Chatbot functions.

The project uses an NLP-based financial chatbot to assist users with questions regarding expenses, salary, budgeting, savings, and financial activities. The overall system is designed to provide intelligent financial assistance while analysing spending patterns and identifying high expenses.

However, the report does not state that reinforcement learning is implemented. Therefore, you should not describe this package as a reinforcement-learning module unless you have actually implemented such an algorithm.

4.2.5 UI PACKAGE

Although the phrase **UI Package** is not explicitly cited in the documentation, the software incorporates user-facing interfaces enabling parents, children, and staff members to interact with financial data and execute operational tasks.

This front-end presentation layer facilitates seamless interaction with tracking mechanisms, allowing users to log transaction details, inspect categorized records alongside graphical analytics, attach digital receipts, audit monthly income versus spending, and engage with the automated assistant. Ultimately, the interface offers a unified, intuitive environment for consolidated personal and institutional resource monitoring.

CHAPTER 5

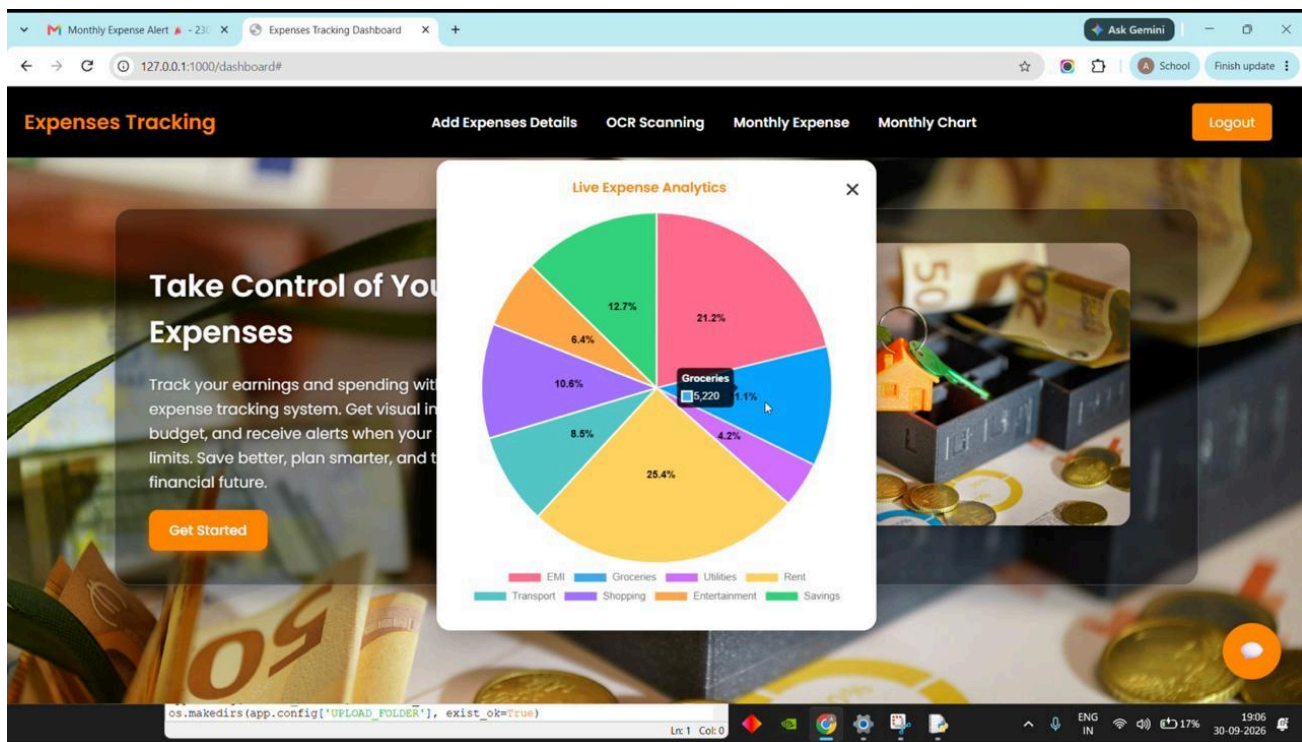
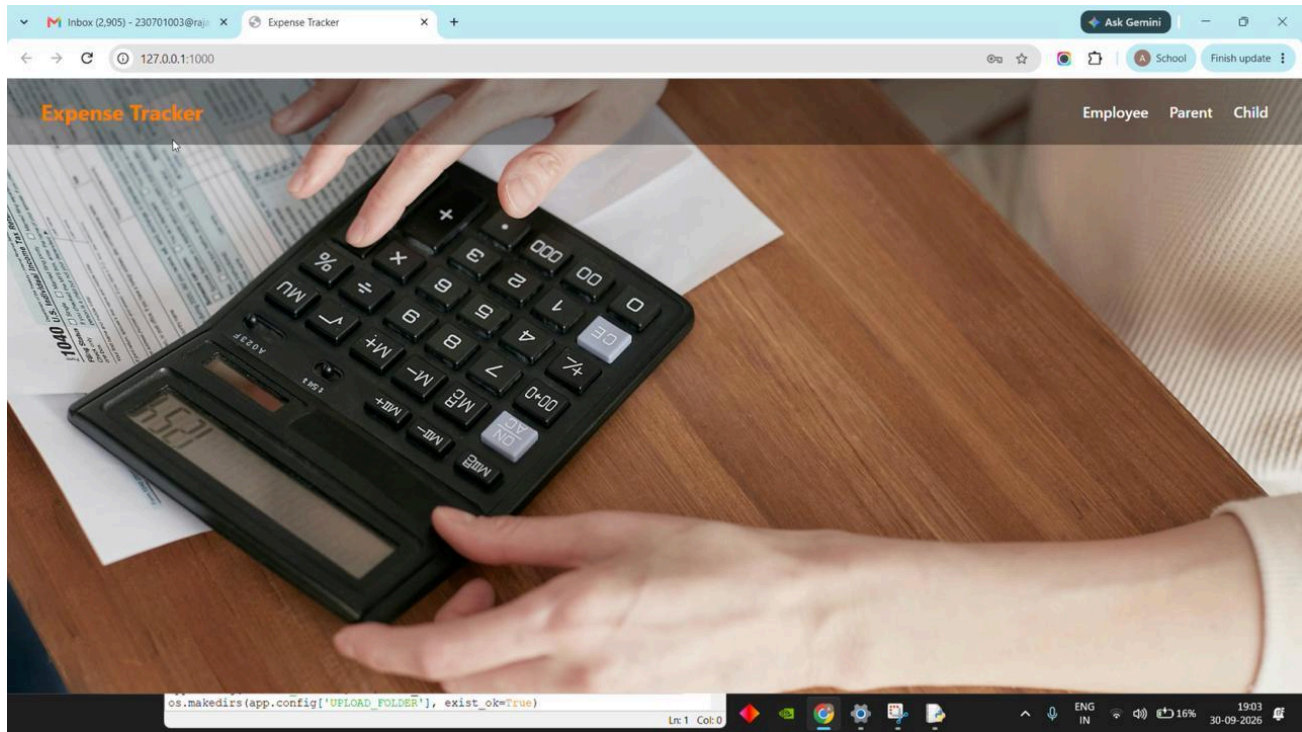
IMPLEMENTATION AND RESULTS

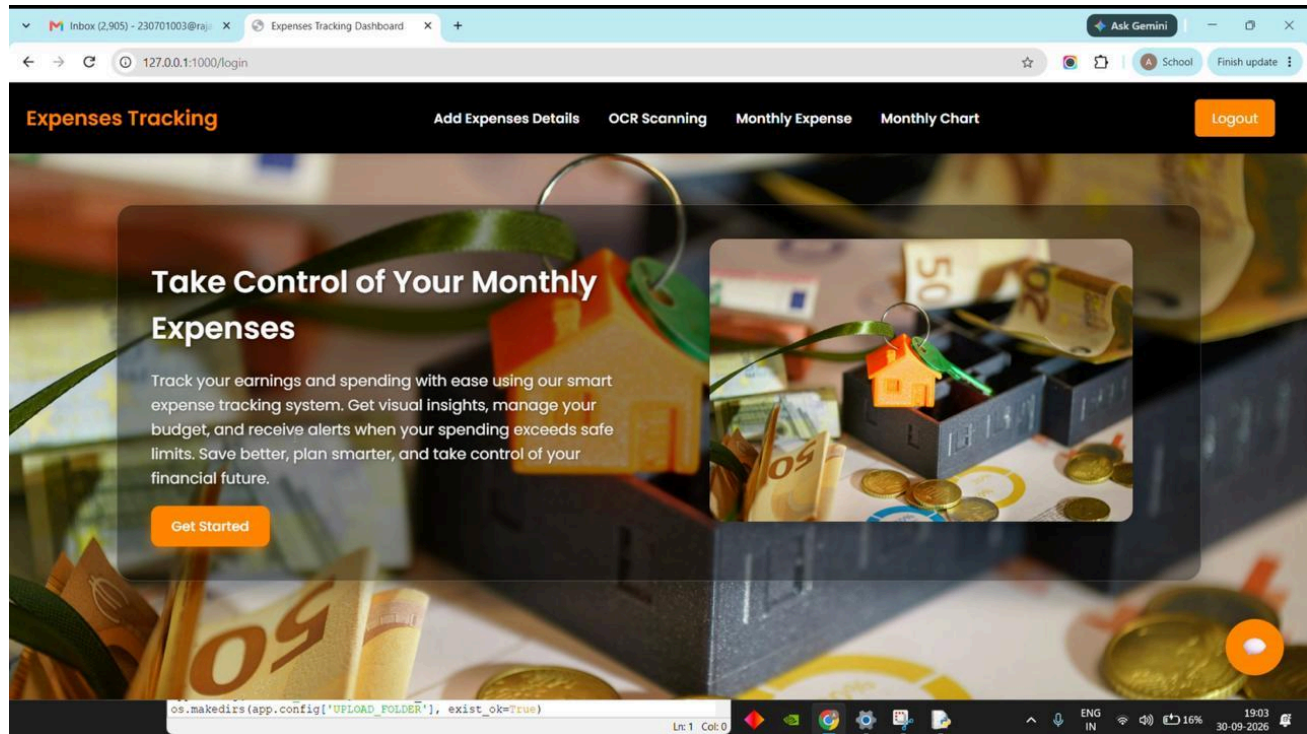
5.1 IMPLEMENTATION RESULTS

The system implementation combines Python-based GUI development, financial-data management, OCR processing, and AI-assisted classification. The available source material explains the implemented techniques and software testing, but it does not provide verified project-specific accuracy values. Therefore, values such as 86.7%, 94.98%, and 94.62% should not be treated as the results of this project because those values belong to the literature-survey papers.

The implementation is organized into several functional modules. These modules include user registration and login, expense and income entry, receipt scanning, expense categorization, financial analysis, graphical reporting, chatbot interaction, and alert generation. Each module contributes to the overall objective of simplifying personal financial management.

The system also supports the storage and retrieval of financial records. Users can add, update, view, and delete transaction details. The stored information can be used to generate summaries based on date, category, amount, or transaction type. This allows users to monitor their financial activities more efficiently.





Expenses Tracking

Add Expenses Details OCR Scanning Monthly Expense Monthly Chart [Logout](#)

Take Control of Your Monthly Expenses

Track your earnings and spending with ease using our smart expense tracking system. Get visual insights, manage your budget, and receive alerts when your spending exceeds safe limits. Save better, plan smarter, and take control of your financial future.

[Get Started](#)

os.makedirs(app.config['UPLOAD_FOLDER'], exist_ok=True)

Ln: 1 Col: 0

127.0.0.1:1000/login

Ask Gemini

School

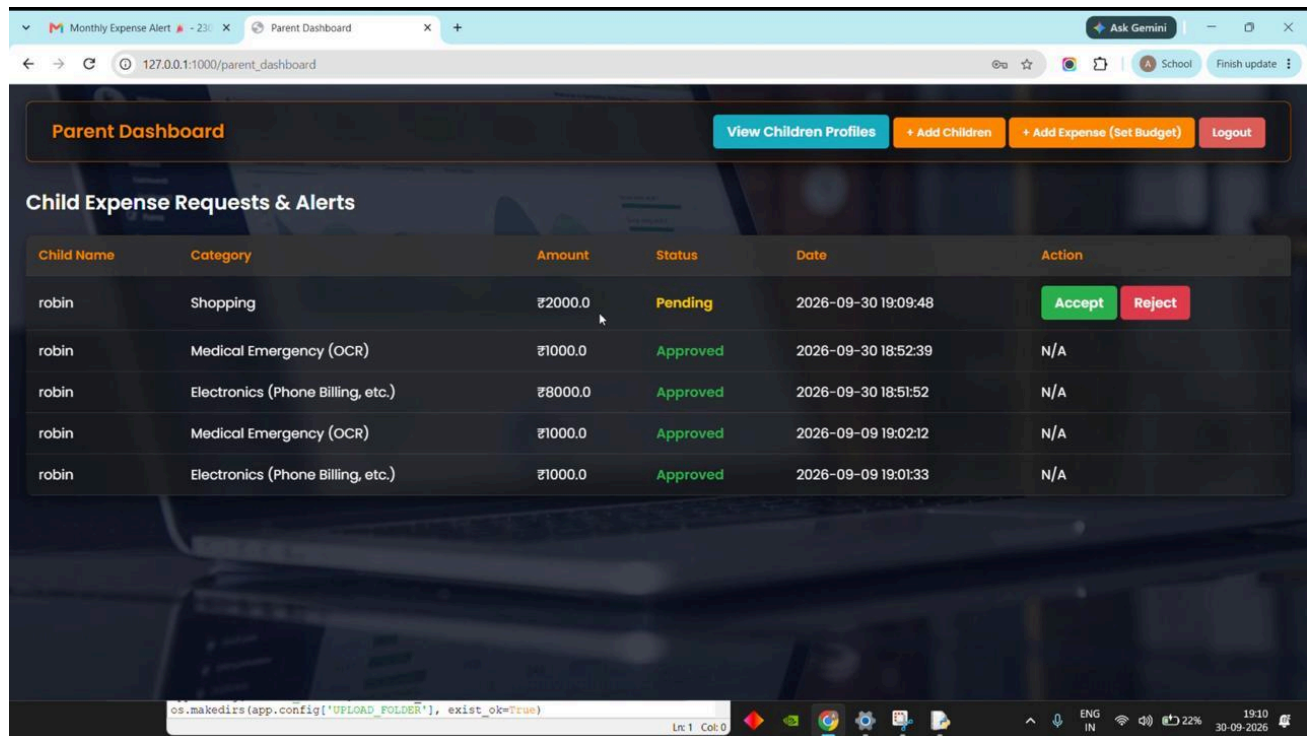
Finish update

ENG IN

16%

19:03

30-09-2026



Parent Dashboard

[View Children Profiles](#) [+ Add Children](#) [+ Add Expense \(Set Budget\)](#) [Logout](#)

Child Expense Requests & Alerts

Child Name	Category	Amount	Status	Date	Action
robin	Shopping	₹2000.0	Pending	2026-09-30 19:09:48	Accept Reject
robin	Medical Emergency (OCR)	₹1000.0	Approved	2026-09-30 18:52:39	N/A
robin	Electronics (Phone Billing, etc.)	₹8000.0	Approved	2026-09-30 18:51:52	N/A
robin	Medical Emergency (OCR)	₹1000.0	Approved	2026-09-09 19:02:12	N/A
robin	Electronics (Phone Billing, etc.)	₹1000.0	Approved	2026-09-09 19:01:33	N/A

os.makedirs(app.config['UPLOAD_FOLDER'], exist_ok=True)

Ln: 1 Col: 0

127.0.0.1:1000/parent_dashboard

Ask Gemini

School

Finish update

ENG IN

22%

19:10

30-09-2026

5.1.1 IMPLEMENTATION OF DATA PRE-PROCESSING

Data preprocessing is used to prepare raw information before it is given to the AI models. For text data, preprocessing includes converting the text into a standard format, removing punctuation, converting words to lowercase, removing unnecessary stop words, cleaning unwanted symbols, and preparing the text for tokenization. The project material identifies removal of stop words, punctuation, and conversion to lowercase as preprocessing operations.

For financial numerical data, the supporting fraud-detection paper explains that raw data may contain noise, missing values, and values in different scales. Therefore, data cleaning, treatment of missing values, normalization, scaling, and dimensionality reduction are performed before model processing.

Additional preprocessing is required for receipt and bill images. The image may be resized, converted into grayscale, and enhanced to improve the visibility of characters. Noise removal, contrast adjustment, thresholding, and image sharpening can help the OCR component recognize printed text more accurately. Proper preprocessing is important because poor-quality images may result in incomplete or incorrect extraction of transaction details.

The preprocessing stage also checks whether the extracted values are valid. For example, the system can verify whether the amount contains a valid numerical value, whether the date follows an acceptable format, and whether the extracted text contains meaningful information. Invalid or incomplete records can be corrected manually before they are stored in the database.

5.1.2 IMPLEMENTATION OF DATA TOKENIZATION

Tokenization is the process of dividing a sentence into individual words or meaningful units so that the system can process the text more easily. For example, the sentence “Show my expenses for this month” can be divided into the tokens “Show”, “my”, “expenses”, “for”, “this”, and “month”. These tokens can then be analysed by the chatbot to identify the user’s intention. The project material identifies tokenization as one of the important Natural Language Processing tasks used in the financial chatbot.

Tokenization also helps the system identify important keywords related to financial activities. Words such as “income”, “expense”, “food”, “transport”, “monthly”, “budget”, and “balance” can be used to determine the type of request made by the user. The chatbot can then connect the identified keywords with the appropriate database query or system function.

In addition, tokenization supports the handling of different forms of user questions. For example, “How much did I spend on food?” and “Show my food expenses” have different sentence structures but contain similar financial meanings. By processing the tokens and identifying the important terms, the chatbot can provide a suitable response to both queries.

5.1.3 IMPLEMENTATION OF DATA EMBEDDING

After tokenization, the words must be converted into numerical representations because machine-learning and deep-learning models cannot directly process normal text. Word embedding converts each word into a numerical vector that represents its meaning in a form suitable for model processing. The project material identifies Word2Vec and GloVe as possible word-embedding approaches. The supporting research paper further explains that Word2Vec creates lower-dimensional dense vector representations and uses approaches such as Continuous Bag of Words and

Skip-Gram. The embedded vectors can then be supplied to CNN or another classification model for further processing.

Embedding helps the system understand the relationship between words and their context. For example, words such as “restaurant”, “grocery”, and “food” may be associated with similar expense categories. Similarly, words such as “salary”, “wages”, and “income” may be related to earnings. This improves the ability of the classification model to assign transactions to suitable categories.

The embedding process also reduces the limitations of simple keyword matching. Instead of depending only on exact words, the model can identify similarities between related terms. This is useful when users enter expenses using different descriptions or when the text extracted from a receipt contains variations in wording.

5.2 DISCUSSION

The proposed system integrates several techniques into a single financial-management platform. OCR reduces the effort required for manually entering bill and receipt information. NLP allows users to interact with the financial chatbot using natural-language queries. CNN-based classification supports automated expense categorization, while the expense-analysis component identifies excessive or unusual spending and generates alerts.

The combination of financial-data management, OCR processing, AI-assisted classification, expense analysis, graphical reporting, and chatbot interaction provides a more intelligent approach than conventional expense-management systems. The system is designed to reduce manual work, improve the organization of financial records, and provide users with better assistance in understanding and monitoring their financial activities.

The system also improves accessibility by allowing users to interact with financial information through both conventional forms and natural-language questions. Users who may not be familiar with complex financial software can enter a simple query and receive information about their balance, expenses, income, or spending categories.

Another important benefit is the integration of automated and manual operations. Although OCR and AI models reduce the amount of repetitive work, users can still review and correct the extracted or predicted information. This provides a balance between automation and reliability.

The system can also support better decision-making by presenting financial information in an organized form. Monthly summaries, category-wise comparisons, budget alerts, and graphical reports help users identify spending habits and plan future expenses. Over time, these features may encourage users to reduce unnecessary spending and improve their saving behaviour.

The performance of the system depends on the quality of the input data. Clear receipt images, complete transaction descriptions, correctly entered amounts, and properly maintained categories can improve the quality of the generated results. In contrast, poor image quality, incomplete records, ambiguous descriptions, and incorrect user input may reduce the effectiveness of OCR, classification, and financial analysis.

However, the available project material does not contain numerical evaluation values for model accuracy, precision, recall, F1-score, OCR accuracy, or chatbot accuracy. Therefore, such values should be included in this chapter only when actual test results obtained from the implemented system are available. Future evaluation should measure the performance of each major component separately, including OCR field-extraction accuracy, expense-classification accuracy, chatbot intent-recognition accuracy, response time, and user satisfaction.

The system can be further improved by adding multilingual chatbot support, voice-based queries, mobile access, automatic bank-statement integration, advanced fraud detection, personalized budget recommendations, and improved receipt recognition. These enhancements could make the platform more flexible and useful for a wider range of users.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

6.1 CONCLUSION

The AI-Based Smart Expense Tracking and Financial Management System provides an intelligent and centralized platform for managing the financial activities of parents, children, and employees. The system brings different financial-management functions into a single application and helps users record, organize, monitor, and analyse their financial information more effectively. It reduces dependence on conventional manual expense-recording methods and provides a structured approach for maintaining daily expenses, salary information, income details, and other financial records.

The system simplifies expense recording by allowing transactions to be entered and categorized according to their purpose. It maintains financial records and provides graphical representations of spending patterns so that users can easily understand how their money is being used. This graphical analysis can help users identify categories in which expenditure is high and thereby improve their awareness of spending behaviour. The system is also designed to support parent-child financial management and employee salary monitoring as part of the same centralized platform.

An important feature of the proposed system is the integration of Optical Character Recognition (OCR) for bill and receipt processing. OCR technology reduces the need for manual bill entry by extracting important information from uploaded bills or

receipts. The extracted data can then be processed by the system and added to the financial records. This improves convenience and reduces the time and effort required for entering transaction information manually.

The system also uses CNN-based classification to assist in the automatic categorization of expenses. By combining OCR-based extraction with classification, the system provides a more automated approach to financial-data entry and organization. This integration is intended to improve the speed of expense recording and support more consistent categorization of transactions.

Another major component is the AI-based high-expense detection mechanism. The system analyses spending information and identifies excessive expenditure. When high spending is detected, an automated alert can be generated and sent to an authorized user. This helps users become aware of unusual or excessive expenditure at an earlier stage and supports better control over financial activities.

The NLP-based financial chatbot adds an interactive component to the system. It allows users to obtain assistance for queries related to expenses, salary, budgeting, and savings. Instead of depending only on conventional menu-based interaction, users can communicate with the system through financial questions. This makes the financial-management application more accessible and user-friendly.

Overall, the proposed system combines expense tracking, graphical financial analysis, salary and financial-record maintenance, OCR-based bill processing, CNN-based expense categorization, high-expense detection, automated alerts, and NLP-based financial assistance. According to the uploaded report, these features are intended to reduce manual effort, help users monitor their finances more efficiently, and improve financial awareness and management.

6.2 FUTURE WORK

The proposed system can be further improved by introducing mobile application support. A mobile version would allow users to access the financial-management functions from different locations and manage their expenses more conveniently. Users could record expenses, upload bills, check financial records, and access the chatbot directly through a mobile device. The uploaded report specifically identifies mobile application integration as an important future enhancement.

Another major future enhancement is the integration of the system with bank accounts and digital payment services. At present, users may need to enter or upload financial information. By connecting the system with banking and digital-payment platforms, transactions could be collected and recorded automatically. This would further reduce manual entry and allow financial records to be updated more efficiently.

The system can also be extended through the use of more advanced Artificial Intelligence and Machine Learning techniques. These techniques could be used to predict future expenses, identify long-term spending trends, and provide personalized budgeting and saving suggestions. Instead of only reporting past expenditure, the system could therefore assist users in planning future financial activities based on their historical spending patterns.

A voice-based financial assistant can also be introduced. This would allow users to record expenses and communicate with the financial chatbot using voice commands rather than typing queries manually. Voice interaction could make the system easier to use, particularly when users need to quickly record financial information or obtain a financial summary.

The future system may also include advanced fraud and suspicious-transaction detection. Such a feature could analyse financial activities and identify transactions or

spending behaviour that appear unusual. When suspicious activity is detected, the system could immediately notify the user or another authorized person. This would strengthen the financial-monitoring and security functions of the application.

Thus, future development can make the system more automated, intelligent, secure, and suitable for long-term financial management. The uploaded report specifically proposes mobile access, banking and digital-payment integration, predictive AI, personalized budgeting and saving suggestions, voice-based assistance, and fraud or suspicious-transaction detection as future enhancements.

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